

Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce

COLLEGE CODE : 8203

COLLEGE : AVC COLLEGE OF ENGINEERING

DEPARTMENT : INFORMATION TECHNOLOGY

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Phase 1: Requirement Analysis & Planning | Phase 2: Salesforce Development – Backend & Configurations

Item	Details
Project Title	Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce
Platform	Skill Wallet (Information Technology, Group project)
Skills Required	Salesforce, Agentforce Automation
Technology	Salesforce Developer Org, Custom Object, Auto-Launched Flow, Agentforce Subagent
Team Lead	Rajkiran R
Team Members	Raja Elamparithi S, Rohith M, Sakthi Balan S, Ragulkumar A
Mentor	Not assigned yet

Phase 1: Requirement Analysis & Planning

Phase 1 covers understanding the problem, the users who face it, the ideas considered, the requirements, and the plan for building the solution.

1.1 Project Overview

Customer support teams receive a large number of tickets every day, but prioritization and assignment are often manual. This leads to delays in resolving critical issues and poor customer satisfaction. This project provides a Salesforce-based solution that analyzes ticket information, classifies urgency, assigns the appropriate support level, and starts automation for critical cases.

The solution uses the Support Ticket Intelligence custom object, an Auto-Launched Flow, and an Agentforce subagent named Support Ticket Priority Analysis. High-priority tickets create an urgent handling task and are assigned to a senior support agent.

1.2 Purpose

- Automatically classify support tickets as High, Medium, or Low priority.
- Retrieve the latest support ticket associated with a customer Account.
- Reduce manual prioritization and assignment effort.
- Create an urgent handling task for High-priority tickets.
- Use Agentforce to analyze ticket details and trigger backend automation.
- Improve ticket response and resolution efficiency.

1.3 Ideation

Problem Statements

No.	I am	I'm trying to	But	Because	Which makes me feel
PS-1	Support Agent	focus on critical tickets first	tickets are not always prioritized automatically	ticket prioritization can be manual	frustrated
PS-2	Customer	get urgent issues resolved quickly	critical issues may wait behind lower-priority tickets	support teams handle large ticket volumes	concerned
PS-3	Support Manager	assign tickets to the right support level	assignment may require manual decisions	ticket urgency is not consistently analyzed	overwhelmed
PS-4	Support Agent	identify urgent issues from descriptions	important keywords can be missed	descriptions contain different urgency signals	uncertain

No.	I am	I'm trying to	But	Because	Which makes me feel
PS-5	Support Manager	reduce SLA-related risk	older unresolved tickets may remain unnoticed	SLA risk needs additional checking	concerned

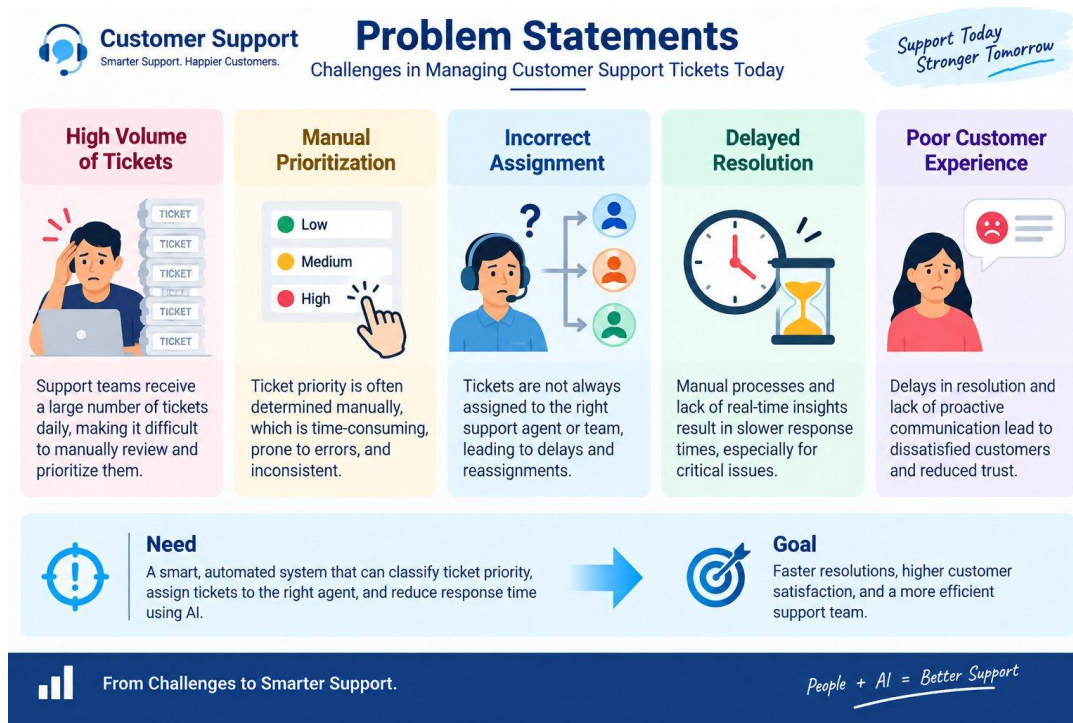


Figure 2: Problem statements – challenges in managing customer support tickets

Empathy Map (Support Agent)

- **Think & Feel:** Wants critical issues identified quickly so urgent customer problems receive attention.
- **Hear:** Receives customer issue descriptions and support requests.
- **See:** Ticket status, issue type, description, priority level, assignment, and SLA risk.
- **Say & Do:** Reviews tickets, handles assigned cases, and focuses on urgent support issues.
- **Pain:** Manual prioritization, missed urgency signals, and delays in handling critical tickets.
- **Gain:** Automatic priority classification, clearer assignment, urgent-task creation, and reduced manual effort.

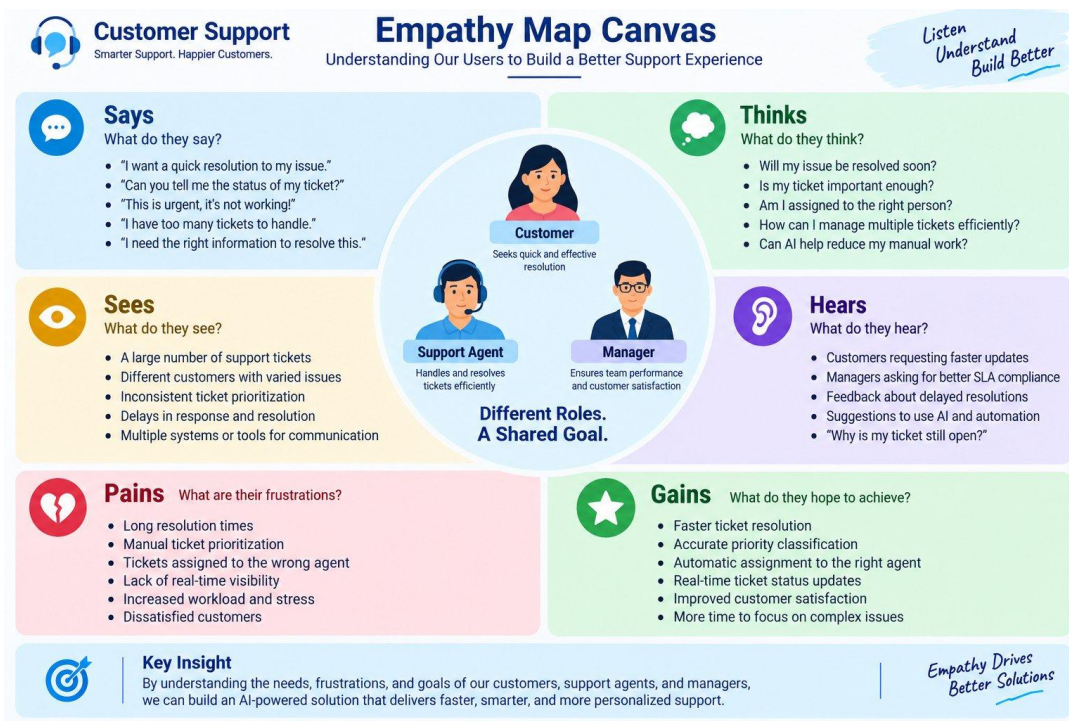


Figure 3: Empathy map canvas

Brainstorming & Idea Prioritization

Ideas considered: manual ticket registers, spreadsheet-based ticket tracking, a standalone support portal, and Salesforce CRM with Agentforce.

Final selection: Salesforce CRM, because it offers:

- Automation through Salesforce Flow
- AI-assisted ticket analysis through Agentforce
- Automatic priority classification
- Automatic urgent-task creation
- Reduced manual support effort

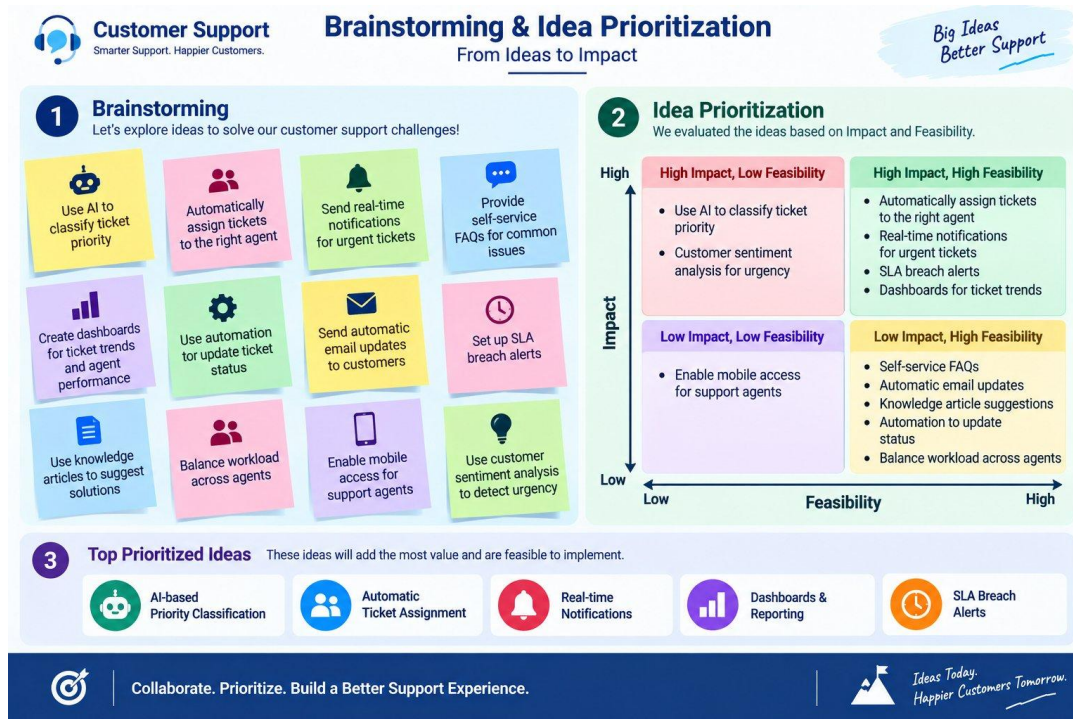


Figure 4: Brainstorming and idea prioritization

1.4 Requirement Analysis

Customer Journey

Ticket Creation → Account Identification → Ticket Retrieval → Description Analysis → Priority Classification → Agent Assignment → Urgent Task Creation → Support Handling → Resolution



Figure 5: Customer journey map

Functional Requirements

FR No	Requirement
FR-1	Support ticket creation and management
FR-2	Account and Contact association
FR-3	Ticket description analysis
FR-4	Automatic priority classification
FR-5	High-priority task creation
FR-6	Support agent assignment
FR-7	SLA breach risk checking
FR-8	Agentforce conversational analysis

Non-Functional Requirements

NFR	Description
Usability	Clear Agentforce interaction and understandable priority messages
Reliability	Consistent priority classification based on configured conditions
Performance	Quick retrieval and analysis of the latest support ticket
Security	Controlled Salesforce access to customer and ticket information
Maintainability	Flow and Agentforce configuration can be updated
Scalability	Supports increasing support-ticket volumes through Salesforce automation

Data Flow

Ticket flow: Customer / Support Request → Account → Latest Support Ticket → Ticket Description → Analyze Description → Priority Level → High/Medium/Low Decision → Agent Assignment → High-Priority Task (if applicable) → Final Response

Agentforce flow: User provides Account Name → Support Ticket Priority Analysis subagent → Auto-Launched Flow → Ticket Analysis → Priority and Assignment Output → Response to User

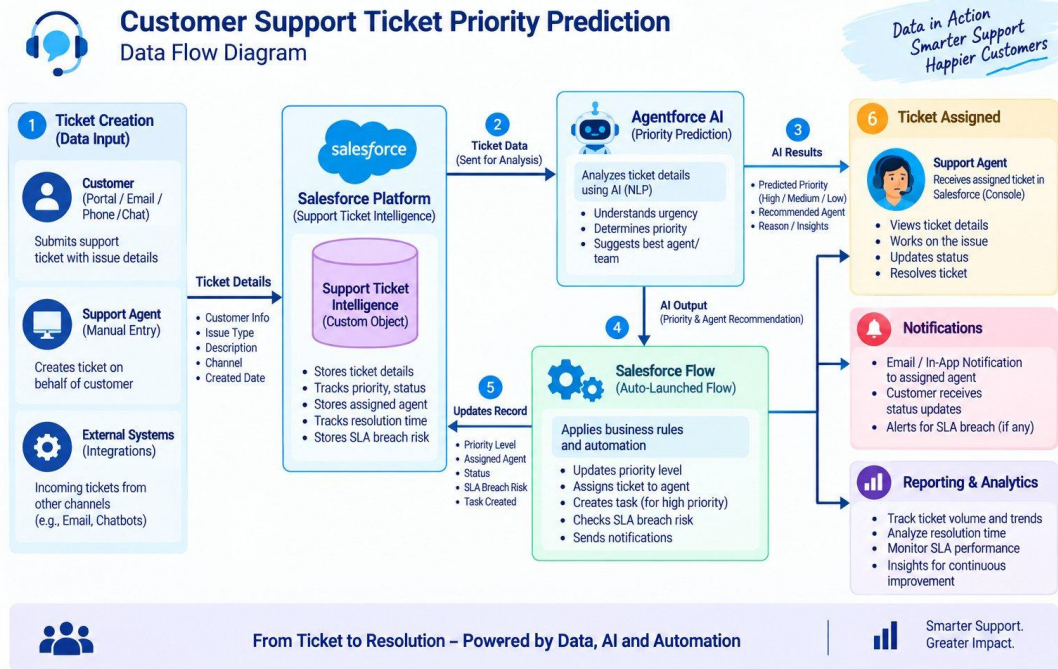


Figure 6: Data flow diagram

Technology Stack

Layer	Technology
Platform	Salesforce
Database	Support_Ticket_Intelligence__c with Account, Contact, User and Task records
Logic	Auto-Launched Flow
AI	Agentforce
Agent Configuration	Support Ticket Priority Analysis subagent
Automation	Flow Decision, Assignment and Create Records
Output	Agentforce conversation response

1.5 Solution Design

Problem–Solution Fit

Problem	Solution	Benefit
Manual ticket prioritization	Flow analyzes description and sets priority	Faster identification of urgent cases
Urgent issues may be missed	High-priority keyword decision	Critical cases receive attention
Manual task creation	Flow creates Task for High priority	Reduced manual effort

Problem	Solution	Benefit
Manual support assignment	Flow sets assigned support level	More consistent handling
SLA risk can be overlooked	Optional SLA Risk decision	Better risk awareness
Need for conversational access	Agentforce subagent	AI-assisted interaction

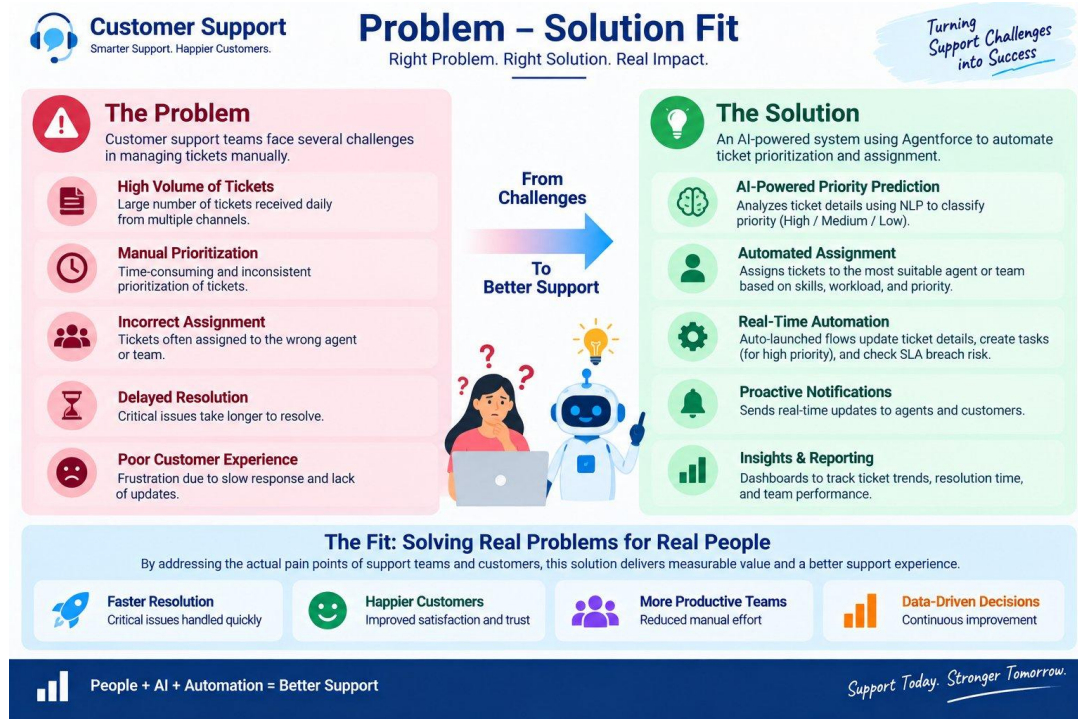


Figure 7: Problem–solution fit

Proposed Solution

Parameter	Description
Problem	Manual ticket prioritization and assignment causing delays in handling critical customer support issues
Solution	Salesforce-based Customer Support Ticket Priority Prediction and Automated Assignment System
Innovation	Agentforce AI + Auto-Launched Flow for analyzing ticket descriptions, automatically classifying priority, assigning support level, and triggering urgent-task automation
Impact	Faster identification of critical tickets, reduced manual effort, improved response time, and better support-team productivity
Scalability	Supports increasing ticket volumes through Salesforce automation, with scope for additional assignment rules, SLA monitoring, and priority indicators

Architecture: Users / Support Team → Agentforce → Support Ticket Priority Analysis → Auto-Launched Flow → Account & Ticket Data → Description Analysis → Priority Decision → Task / Assignment → Final Message → Agentforce Response

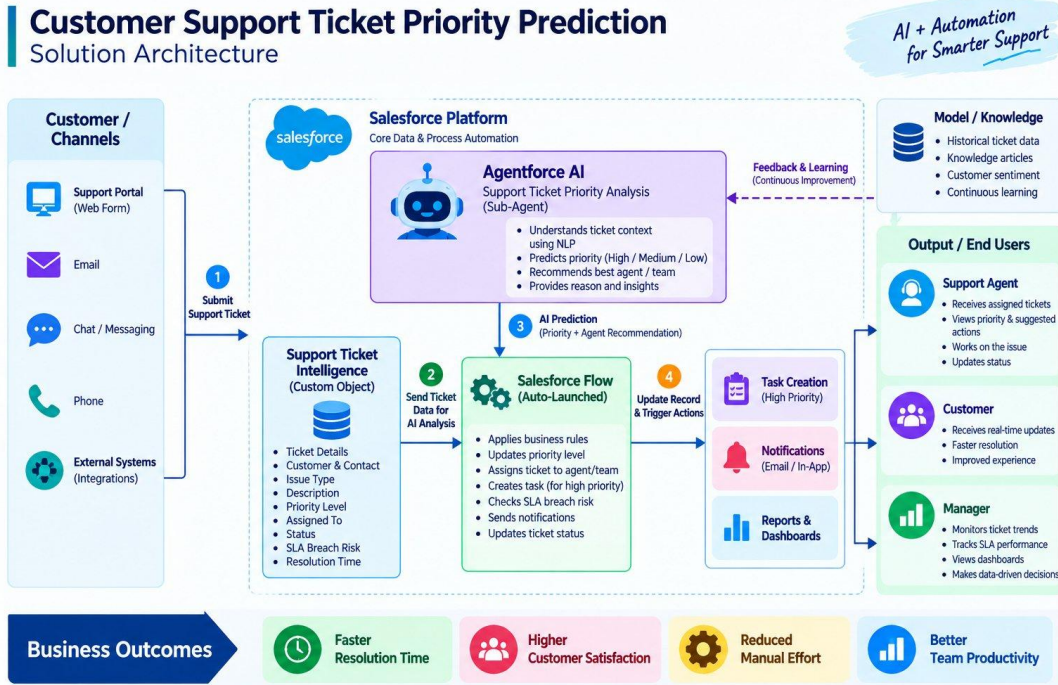


Figure 8: Solution architecture

1.6 Project Planning

The project follows an Agile approach: epics are broken into stories with story points, work is executed in sprints, and velocity is used for estimation. The Skill Wallet project tracker (Figure 1) shows 2 epics, 8 tasks and 2 subtasks.

Sprint	Epic	USN	User Story / Task	Points	Priority
1	Developer Setup	USN-1	As a system administrator, I want to configure the Salesforce environment for Support Ticket Intelligence so that the support automation solution can be developed.	3	High
2	Data Modeling	USN-2	As an admin, I want to create the Support Ticket Intelligence object with required fields and relationships so that ticket information is stored systematically.	5	High
2	Data Modeling	USN-3	As a support user, I want tickets associated with Accounts and Contacts so that the latest customer ticket can be identified for analysis.	3	High
3	Automation	USN-4	As a support agent, I want an Auto-Launched Flow to retrieve the latest ticket and analyze its description so that priority can be determined automatically.	5	High

Sprint	Epic	USN	User Story / Task	Points	Priority
3	Automation	USN-5	As a support agent, I want tickets classified as High, Medium, or Low based on configured urgency conditions so that critical issues receive attention first.	5	High
4	Agentforce & Automation	USN-6	As an AI Agent (Agentforce), I want to analyze ticket details and trigger the backend Flow so that manual prioritization is reduced.	8	High
4	Task & Assignment	USN-7	As a support manager, I want High-priority tickets to create an urgent handling task and assign the appropriate support level so that critical tickets are handled quickly.	5	High
5	SLA Management	USN-8	As a support manager, I want the system to check SLA breach risk for older unresolved tickets so that potential SLA issues can be identified.	5	Medium
6	Agentforce & Conversational Support	USN-9	As a support user, I want to provide an Account Name to Agentforce and receive the ticket ID, priority, assigned agent, and action message so that I understand the outcome conversationally.	5	Medium

Phase 2: Salesforce Development – Backend & Configurations

Phase 2 covers building the solution in Salesforce: the developer org, the data model, the Auto-Launched Flow, and the Agentforce subagent and action, followed by testing.

2.1 Milestone 1: Salesforce Developer Org

Activity 1: Creating a Developer Account

1. Go to <https://developer.salesforce.com/signup>.
2. Enter first and last name, email, Role (Developer), Company (college name), Country (India), Postal Code, and a Username (a combination of your name and company, in the format username@organization.com).
3. Click Sign me up.

Activity 2: Account Activation

1. Open the verification email (it may take 5–10 minutes) and click Verify Account.
2. Set a password, answer a security question, and click Change Password.
3. You are redirected to the Salesforce Setup page.

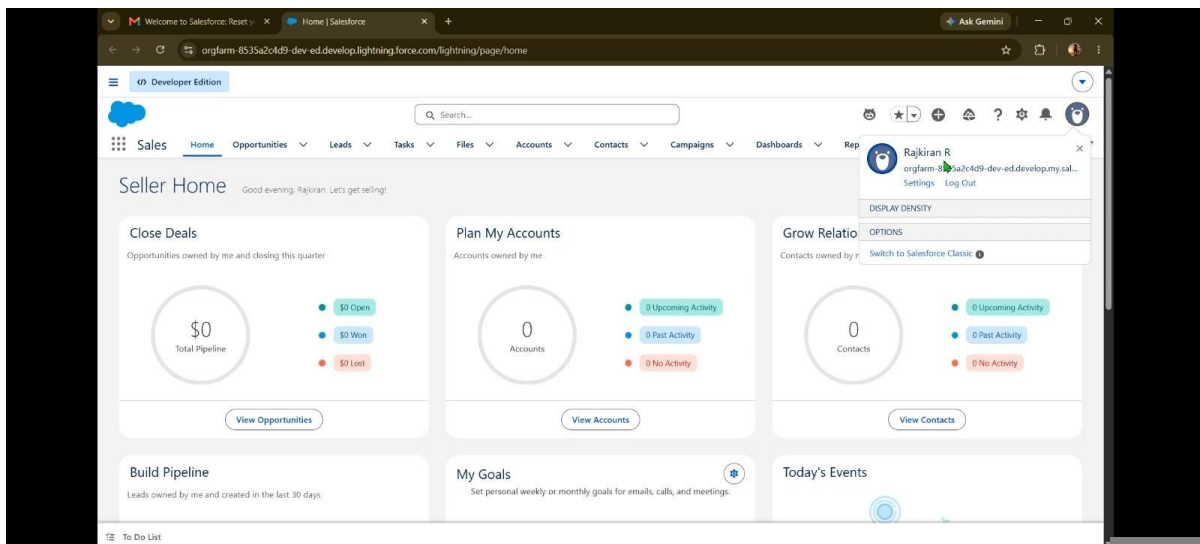


Figure 9: Salesforce Developer Edition org – Sales app home page

2.2 Milestone 2: Data Management – Objects

Custom object: Support Ticket Intelligence **API name:** Support_Ticket_Intelligence__c

Field Label	API Name	Data Type	Description
Ticket Number	Ticket_Number__c	Auto Number	TKT-{0000}
Customer	Customer__c	Lookup (Account)	Related Account
Contact	Contact__c	Lookup (Contact)	Customer contact

Field Label	API Name	Data Type	Description
Issue Type	Issue_Type__c	Picklist	Technical, Billing, General
Description	Description__c	Long Text Area	Issue details
Priority Level	Priority_Level__c	Picklist	Low, Medium, High
Status	Status__c	Picklist	New, In Progress, Resolved
Created Date	Created_Date__c	Date	Ticket date
Assigned To	Assigned_To__c	Lookup (User)	Support agent
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	Risk flag
Resolution Time (hrs)	Resolution_Time__c	Number	Time taken

A test ticket was created from the New Support Ticket Intelligence form, linking a Customer (Account) and a Contact, selecting the Issue Type, and choosing the Priority Level from the picklist (Low, Medium, High).

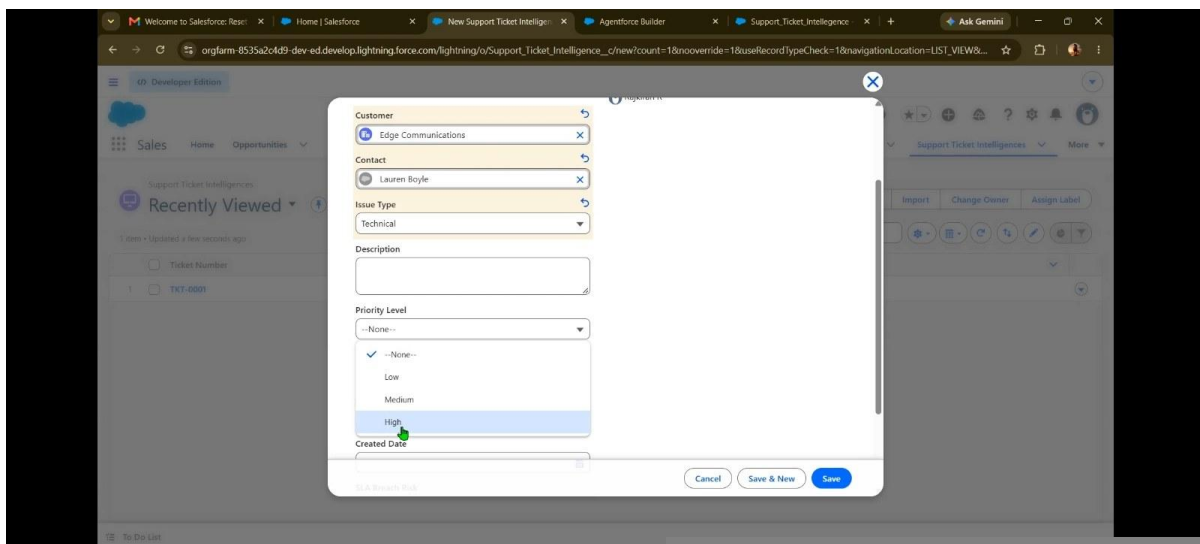


Figure 10: Creating a Support Ticket Intelligence record – Customer, Contact, Issue Type and Priority Level

2.3 Auto-Launched Flow

Flow type: Auto-Launched Flow (no trigger; called from Agentforce). **Flow name:** Support_Ticket_Intelligence

Variables

Label	API Name	Type	Availability
Account Name	varAccountName	Text	Input
Account Id	varAccountId	Text	Output
Ticket Id	varTicketId	Text	Output

Label	API Name	Type	Availability
Priority Level	varPriorityLevel	Text	Output
Assigned Agent	varAssignedTo	Text	Output
Action Message	varActionMessage	Text	Output

Flow Steps

1. **Get Account (Get Records):** Account where Name equals {!varAccountName}; sort by CreatedDate descending; store only the first record.
2. **Store Account Id (Assignment):** varAccountId = {!Get_Account.Id}.
3. **Get Ticket (Get Records):** Support_Ticket_Intelligence__c where Customer__c equals {!varAccountId}; sort by CreatedDate descending; store only the first record.
4. **Store Ticket Id (Assignment):** varTicketId = {!Get_Ticket.Id}.
5. **Analyze Description (Decision):** High if the description contains "urgent", "not working" or "failure"; Medium if it contains "issue", "slow" or "delay"; default outcome is Low.
6. **Set Priority Level (Assignment):** varPriorityLevel = "High", "Medium" or "Low" on the matching path.
7. **Is High Priority (Decision):** checks whether varPriorityLevel equals "High".
8. **Create Task (Create Records, High only):** Subject "Urgent Ticket Handling", WhatId {!varTicketId}, Priority High, Status Not Started.
9. **Set Assigned Agent (Assignment):** varAssignedTo = "Senior Support Agent".
10. **Optional SLA check (Decision):** flags tickets whose Created Date is older than 2 days.
11. **Set Final Message (Assignment):** High – "High priority ticket detected. Assigned to senior agent."; Medium – "Ticket marked as medium priority. Will be handled shortly."; Low – "Tickets are low priority and queued for processing."
12. **Save and activate the flow.**

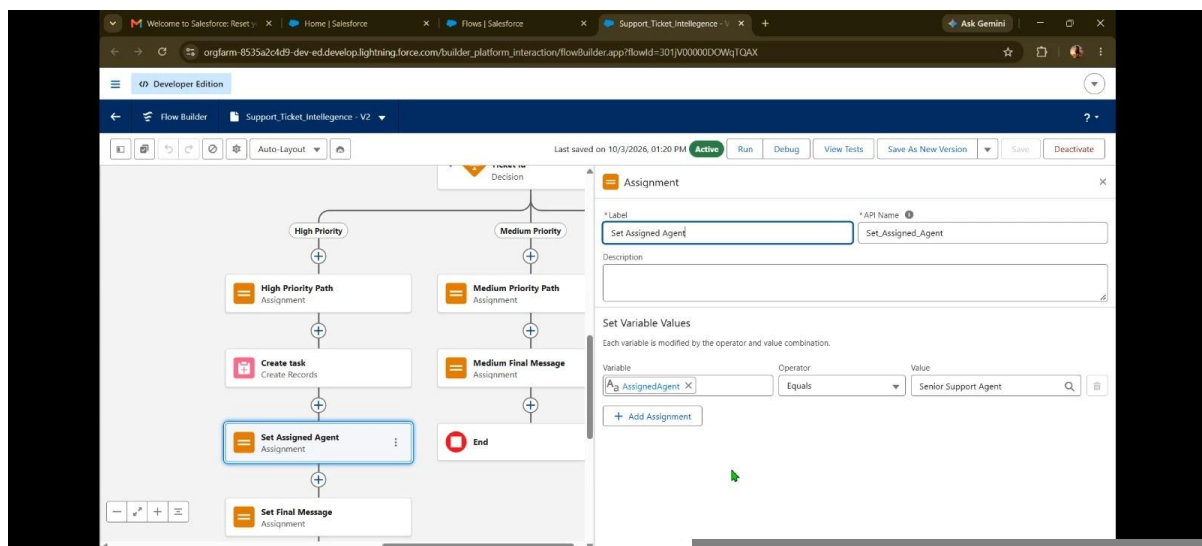


Figure 11: Flow Builder – High Priority path with Create Task and the Set Assigned Agent assignment (Senior Support Agent); flow is Active

2.4 Agentforce Configuration

Enable Agentforce

1. Click the gear icon and choose Salesforce Go.
2. Search for "default", select Agentforce (Default), and enable it (Get Started → Turn On → Confirm).
3. In Setup, search for Agents in Quick Find, enable the agent, and open Agentforce (Default) in Builder.

Create the Subagent

Property	Value
Name	Support Ticket Priority Analysis
API Name	Support_Ticket_Priority_Analysis
Classification Description	Analyzes customer support tickets to determine priority based on issue description and urgency. Handles user requests related to ticket status, issue severity, and support prioritization.
Scope	Only analyze support ticket descriptions, determine the priority level (High, Medium, Low), and trigger backend automation for task assignment. Do not handle unrelated requests such as billing, subscription management, or account updates.

Subagent Instructions

1. Ask the user to provide the Account Name if not already given.
2. Retrieve the latest support ticket associated with the Account.
3. Read and analyze the ticket description.
4. Determine priority from keywords – High: "urgent", "not working", "failure"; Medium: "issue", "slow", "delay"; Low: none of the above.
5. Set the priority level accordingly.
6. If the ticket is High priority, trigger the Auto-Launched Flow and ensure a task is created for immediate handling.
7. Assign the ticket to the appropriate support level (for example, senior agent for high priority).
8. Return a clear response – High: urgent handling is initiated; Medium: issue will be handled shortly; Low: ticket is queued.
9. Do not update records directly or send emails. Use Flow for all actions.

Agent Action (Flow-based)

Reference action type: Flow – Support_Ticket_Intelligence__c. **Description:** Analyzes customer support tickets based on the provided account name, determines ticket priority using the issue description, and triggers automation to assign tasks for high-priority cases. **Loading text:** "Analyzing ticket details and determining priority..."

Variable	Direction	Settings	Description
varAccountName	Input	Require input; collect data from user	Customer account name used to fetch the latest ticket
varAccountId	Output	Show in conversation; text	Account identifier used to retrieve the ticket
varTicketId	Output	Show in conversation; text	Identifier of the ticket analyzed
varPriorityLevel	Output	Show in conversation; text	Calculated priority (High, Medium, Low)
varAssignedTo	Output	Show in conversation; text	Agent or team assigned to the ticket
varActionMessage	Output	Show in conversation; text	Final message with priority and actions taken

2.5 Testing in Agentforce Builder

The agent was tested in the Conversation Preview of Agentforce Builder (Agentforce Default, Version 1, Active). Asking "give me my ticket details" makes the agent ask for the AccountName input; entering a value and submitting it runs the Flow action.

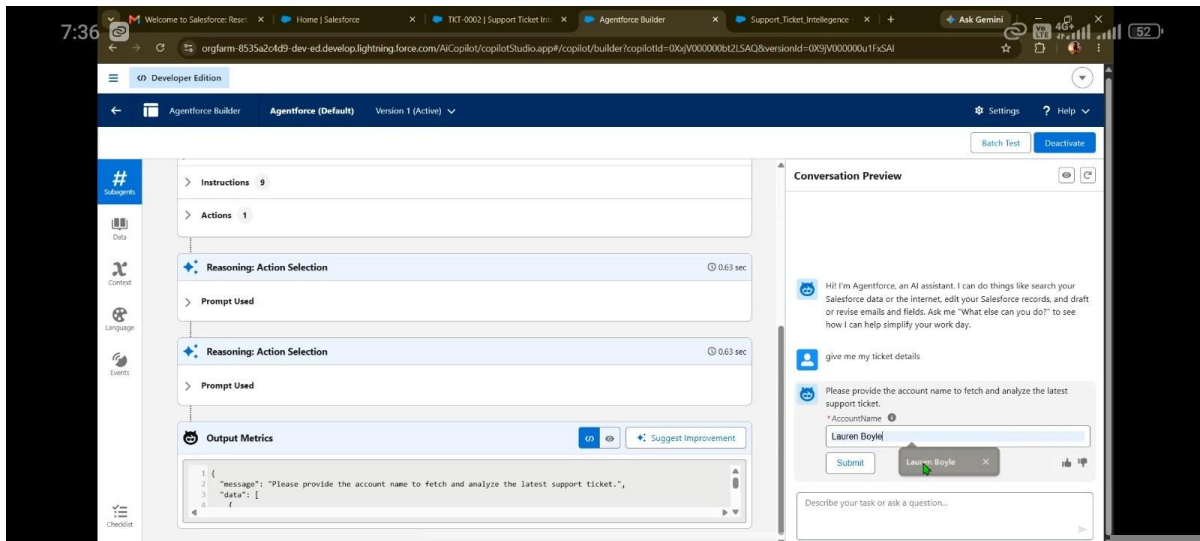


Figure 12: Conversation Preview – the agent asks for the AccountName and the value is entered

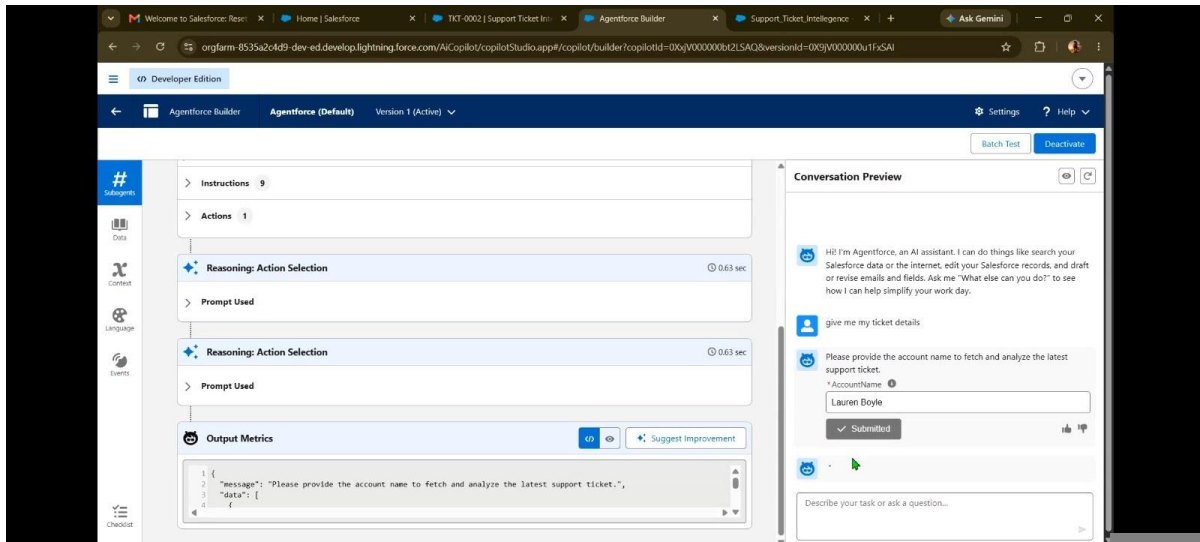


Figure 13: Conversation Preview – AccountName submitted and the action running; reasoning steps and output metrics shown on the left

2.6 Outcome of Phase 2

- Takes Account Name as input.
- Fetches the latest support ticket for that Account.
- Analyzes the description using keywords and sets the priority automatically.
- Creates an "Urgent Ticket Handling" task for High-priority tickets.
- Returns the ticket ID, priority, assigned agent, and action message to Agentforce.